Automated Extraction of Fever Temporal Relations from EHR Discharge Notes

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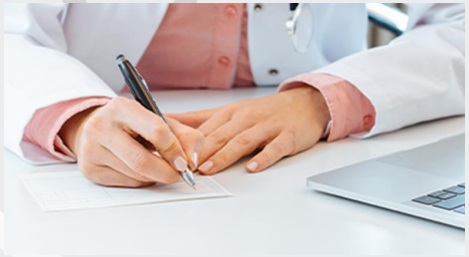
Project Goal: Develop an automated pipeline that identifies and extracts all instances of **"fever" phenotypes** along with their **associated temporal relation expressions** (eg. onset, duration) from **unstructured narrative notes in the MIMIC database**, producing categorized outputs for downstream disease trajectory modelling and personalized medicine

Project Approach:

- Unstructured Notes: MIMIC IV Notes – Patient Discharge Summary
- LLM Model: Meta Llama 3.2 70B Instruct model
- Fever phenotype detection with PhenoBERT (BERT-based clinical NLP model)
- Negation filtering (NegEx), eg. 'no fever'
- LLM Prompt engineering
- Temporal relation expression mining with Llama model

Project Steps

Clinical Discharge Notes



PhenoBERT

Sample Output:

295	300	fever	HP:0001945	1.00
509	515	fevers	HP:0001945	1.00
663	668	fever	HP:0001945	1.00
1220	1225	fever	HP:0001945	1.00

NegEx

"Fever" at position 663-668 is **negated**, removed

LLM Temporal Relation Extraction

Inputs: Character Positions, Entire Note

Prompt: One-shot (example)

Output: Temporal Relation Expression (free text)

Standardization

Fever Mention	Time Expression	Relation	Related Symptom/Event	Clarification
Fever onset	"yesterday"	BEGINS	–	Fever started yesterday
Fever	"for 3 days"	DURING	Headache	Fever overlapped headache
Fever	"after surgery"	AFTER	Surgery	Fever appeared post-op
Fever resolution	"this morning"	ENDS	–	Fever ended this morning
Fever episode	"days 2-4"	OVERLAP	Rash	Symptoms coincided

Current Prompt Design

run_phenobert.py U | hpo_pheno | test_openrouter.py | temp.csv | notes_dbmi.csv

Users > Selina > Documents > GitHub > test_openrouter.py > ...

```python
from openai import OpenAI

client = OpenAI(
  base_url="https://openrouter.ai/api/v1",
  api_key="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",
)

input_text = (
    "<|begin_of_text|><|start_header_id|>system<|end_header_id|>\n"
    "You are a clinical informatics expert. We detected the HPO phenotype term 'fever' from the clinical discharge narrative note."
    "Identify all time periods and points when 'fever' is reported, using relations such as BEFORE, AFTER, DURING, OVERLAP, BEGINS, or ENDS.\n\n"
    "Make sure to only report time points and connections between symptoms and the 'fever' phenotype, including imprecise, relative, and compound time expressions.
    "Example input: The patient reported a high fever beginning midday, which resolved by midnight.\n\n"
    "Example output: Fever onset began at midday. Fever ended by midnight. Fever interval was from midday to midnight. No related clinical events.\n\n"
    "<|start_header_id|>user<|end_header_id|>\n"
    "As such, Ms. ___ was admitted for a work-up of fever and white count, and for treatment of recurrent ascites."
    "<|eot_id|><|start_header_id|>assistant<|end_header_id|>"
)
completion = client.chat.completions.create(model="meta-llama/llama-3.3-70b-instruct:free",
  messages=[
    {
      "role": "user",
      "content": input_text
    }
  ]
)

print(completion.choices[0].message.content)
```

Prompt

One-shot

Current Progress

- MIMIC notes data access, cleaning, and preprocessing
- Literature review on phenotype entity recognition, temporal reasoning
- Narrow down to ‘fever’ phenotype and temporal relation extraction
- Tried models: PhenoBERT, NegEx package, PhenoGPT2, NegBERT
- Prompt engineering: reference Meta Llama recommended Instruct format
- Researched on specific temporal intervals in AI algorithms: BEFORE, AFTER, DURING, OVERLAP, BEGINS, and ENDS

Challenges & Next Steps

- **Very large CSV** discharge note data, takes long time to run through everything
  - Processing by chunk (per row) improves efficiency
  - Alternatives: Multiprocessing, *polars* library
- **Messy, inconsistent discharge notes packed with jargon + abbreviations**
  - Difficult for LLM to process generalizations and context
  - Requires preprocessing, fine-tuning/training model, post-processing
- **Insufficient memory/CPU** to run Meta Llama 3B on local drive
  - Tried Google Colab (limited memory)
  - Use cloud-based OpenRouter, more free memory
- **Next Steps:** Standardize the output format of temporal relation extraction and categorize into temporal intervals